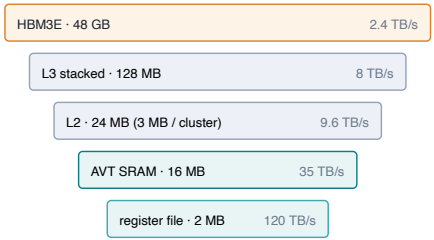
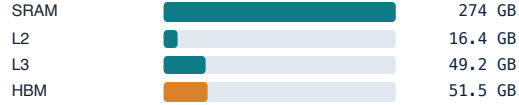
a Memory hierarchy



The working set of a tile (2 time levels × 24 variables × 910 cells) fits in 256 KB

b Bytes moved per time step



The tile SRAM sees 5.3× more traffic than HBM and costs 213× less per byte, which is why the fused pass is the central design choice.

c Energy per byte

level	pJ/byte	relative
HBM3E	`16.0`	213x
L3	`0.40`	5.3x
L2	`0.138`	1.8x
AVT SRAM	`0.075`	1.0x

A halo byte that leaks to L3 costs 5.3 tile-SRAM bytes; one that leaks to HBM costs 213. The fabric exists to keep halos in the first column.

d Residency rule of thumb

256 KB tile SRAM
÷ (24 variables × 4 B × 3 time levels) = **910 cells**

64 tiles → **58 k cells** in flight
4.19 M cells → 72 sweeps per step

state streamed to HBM once per step = 51.5 GB

Larger tiles would cut HBM traffic but exceed the SRAM energy budget; 910 cells is the optimum of the model.